SPINNAKER ANALYTICS

Predicting Household Appliances Energy Consumption

Bhoday Jasmeet Singh
Data Science & AI
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AGENDA

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- Exploratory Data Analysis (EDA)
- Preprocessing and Feature Engineering
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- Key Insights and Applications
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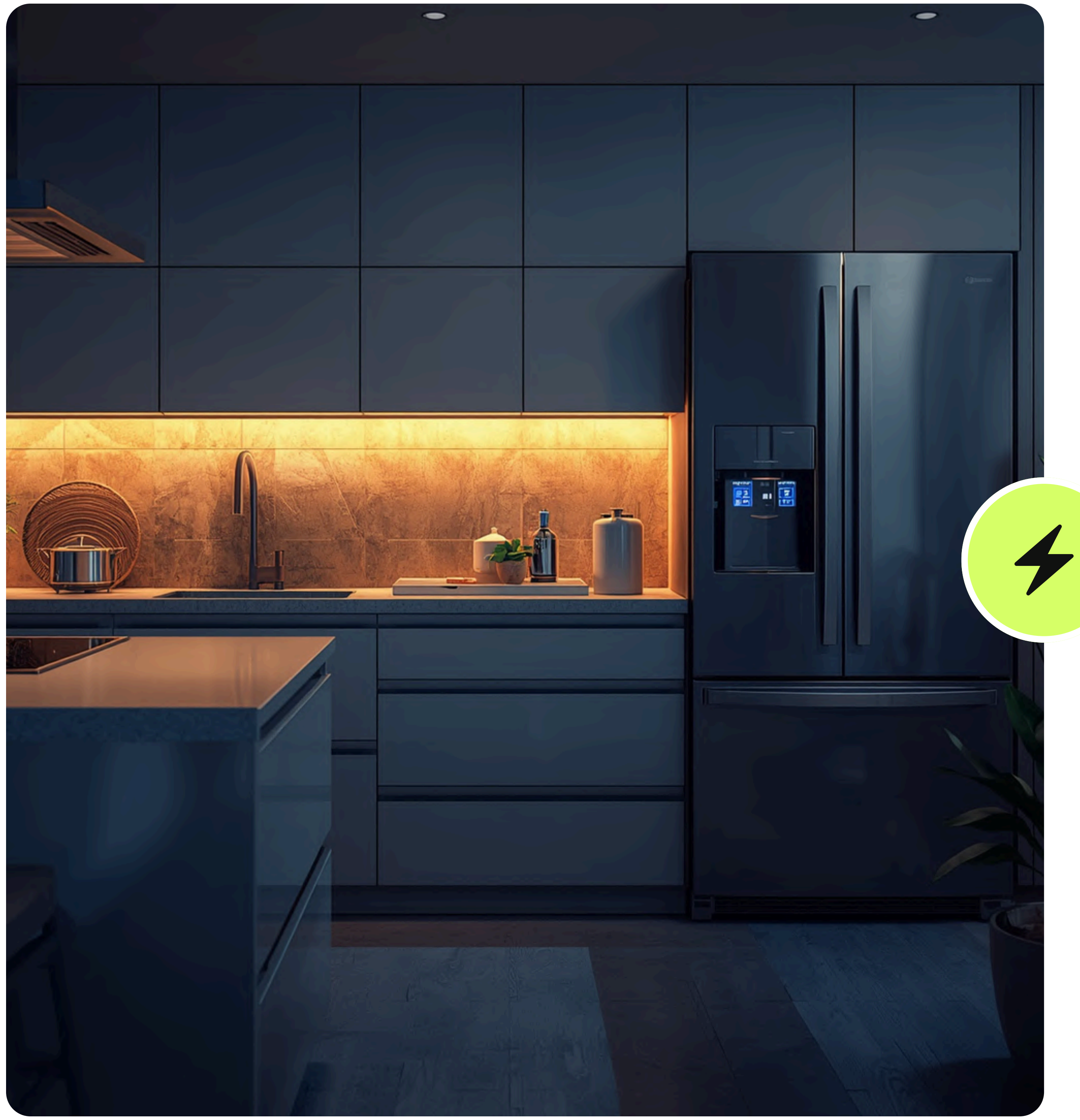
Introduction



- Energy consumption prediction plays a crucial role in modern smart systems, especially in smart homes and energy-efficient environments.
- It helps optimize electricity usage, reduce costs, and improve sustainability.
- Machine learning techniques are highly effective in this domain as they can analyze large amounts of historical data and identify complex patterns in energy usage.
- This study leverages data-driven approaches to improve prediction accuracy.

PROBLEM STATEMENT

- The goal of this project is to predict appliance energy consumption (in Wh) using various environmental and indoor factors such as temperature, humidity, and weather conditions.
- Traditional statistical methods struggle to capture the complex and non-linear relationships between these variables.
- Therefore, machine learning models are used to better understand these interactions and generate more accurate predictions.

OBJECTIVES

- The main objective of this project is to analyze the dataset and understand the factors affecting appliance energy consumption.
- It involves performing data preprocessing, applying suitable machine learning algorithms, and comparing their performance.
- The final goal is to identify the best-performing model that can accurately predict energy consumption based on the given features.

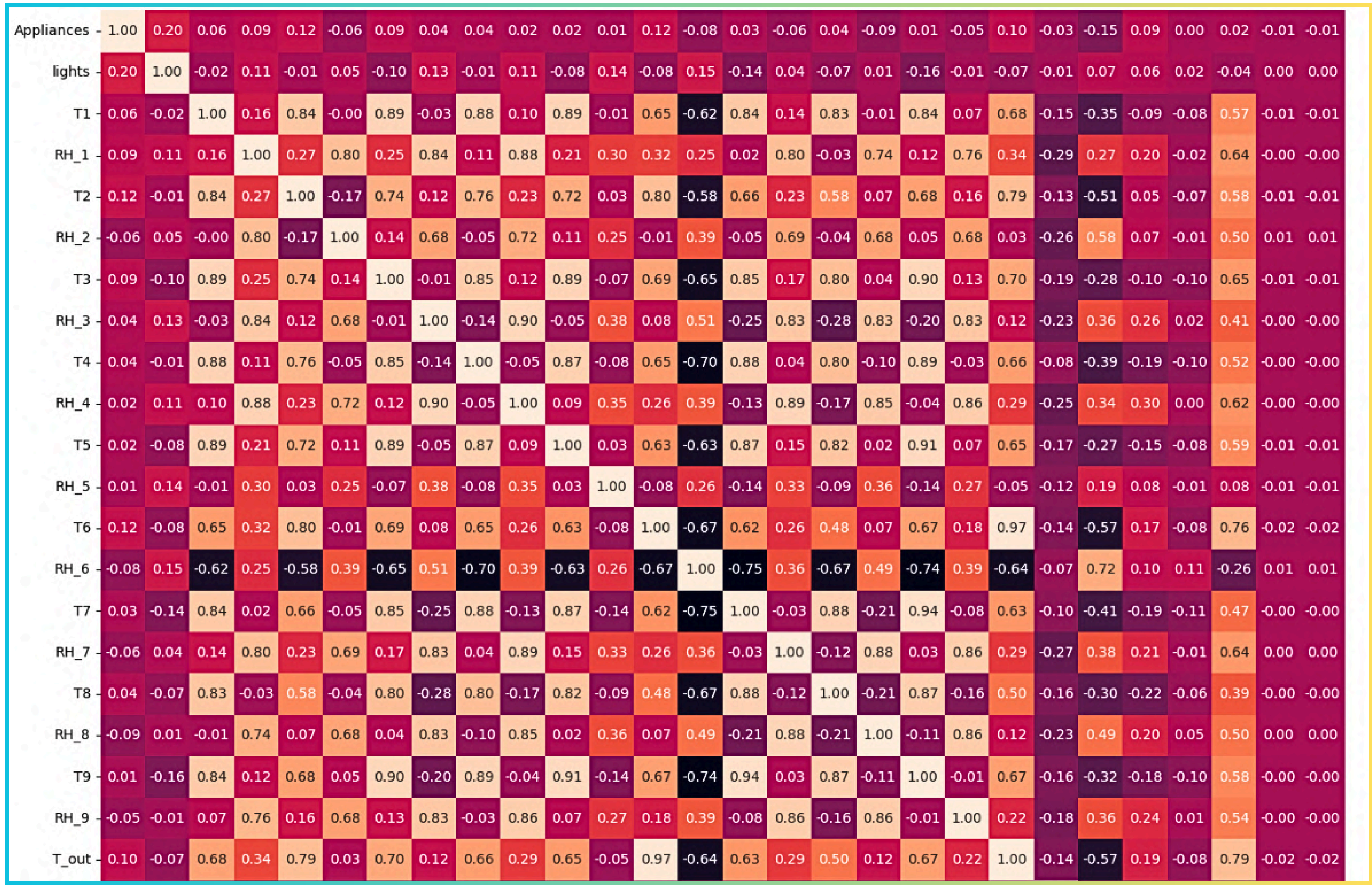
DATASET OVERVIEW

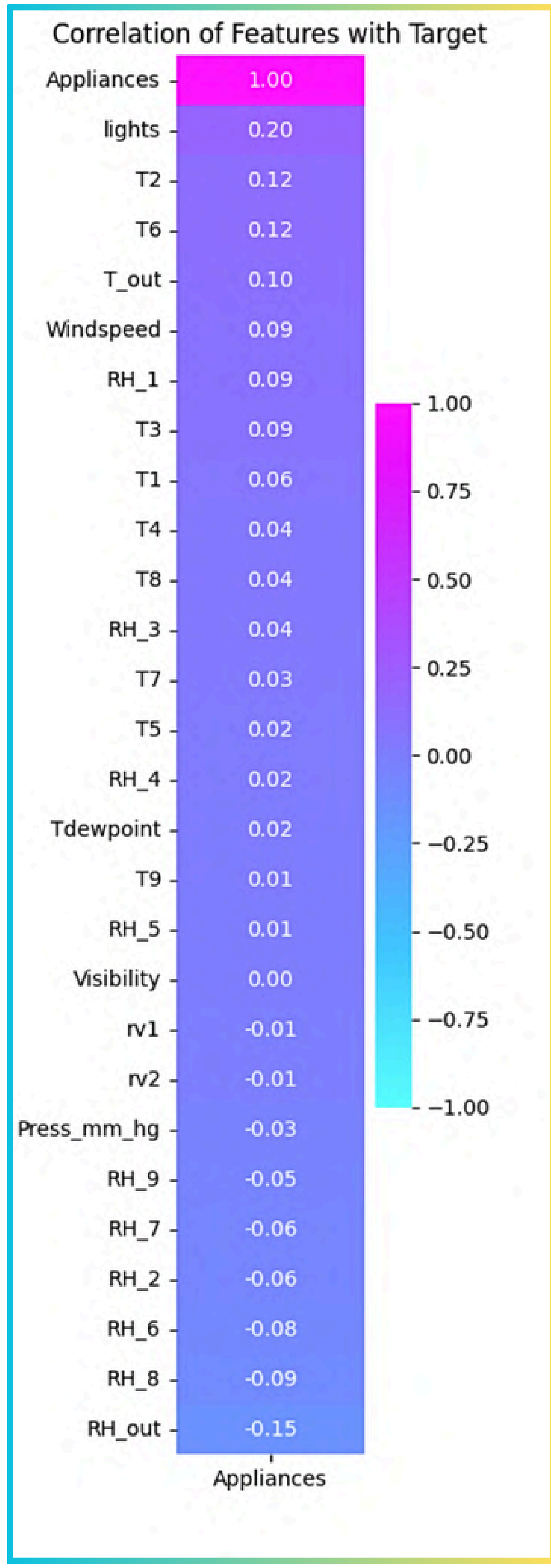
- The dataset used in this project contains 19,735 rows and 29 columns, representing various environmental and indoor conditions affecting energy consumption.
- The target variable is “Appliances”, which indicates the energy usage in watt-hours (Wh).
- Most of the features are numerical and the dataset does not contain missing or duplicate values, making it suitable for analysis and modeling.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
date	Appliance	lights	T1	RH_1	T2	RH_2	T3	RH_3	T4	RH_4	T5	RH_5	T6	RH_6	T7	RH_7	T8	RH_8
11-01-2016 17:00	60	30	19.89	47.59667	19.2	44.79	19.79	44.73	19	45.56667	17.16667	55.2	7.026667	84.25667	17.2	41.62667	18.2	48.9
11-01-2016 17:10	60	30	19.89	46.69333	19.2	44.7225	19.79	44.79	19	45.9925	17.16667	55.2	6.833333	84.06333	17.2	41.56	18.2	48.86333
11-01-2016 17:20	50	30	19.89	46.3	19.2	44.62667	19.79	44.93333	18.92667	45.89	17.16667	55.09	6.56	83.15667	17.2	41.43333	18.2	48.73
11-01-2016 17:30	50	40	19.89	46.06667	19.2	44.59	19.79	45	18.89	45.72333	17.16667	55.09	6.433333	83.42333	17.13333	41.29	18.1	48.59
11-01-2016 17:40	60	40	19.89	46.33333	19.2	44.53	19.79	45	18.89	45.53	17.2	55.09	6.366667	84.89333	17.2	41.23	18.1	48.59
11-01-2016 17:50	50	40	19.89	46.02667	19.2	44.5	19.79	44.93333	18.89	45.73	17.13333	55.03	6.3	85.76667	17.13333	41.26	18.1	48.59
11-01-2016 18:00	60	50	19.89	45.76667	19.2	44.5	19.79	44.9	18.89	45.79	17.1	54.96667	6.263333	86.09	17.13333	41.2	18.1	48.59
11-01-2016 18:10	60	50	19.85667	45.56	19.2	44.5	19.73	44.9	18.89	45.86333	17.1	54.9	6.19	86.42333	17.1	41.2	18.1	48.59
11-01-2016 18:20	60	40	19.79	45.5975	19.2	44.43333	19.73	44.79	18.89	45.79	17.16667	55	6.123333	87.22667	17.16667	41.4	18.1	48.59
11-01-2016 18:30	70	40	19.85667	46.09	19.23	44.4	19.79	44.86333	18.89	46.09667	17.1	55	6.19	87.62667	17.2	41.5	18.1	48.59
11-01-2016 18:40	230	70	19.92667	45.86333	19.35667	44.4	19.79	44.9	18.89	46.43	17.1	55	6.19	87.86667	17.2475	42.7175	18.1	48.59
11-01-2016 18:50	580	60	20.06667	46.39667	19.42667	44.4	19.79	44.82667	19	46.43	17.1	55	6.123333	87.99333	17.53	44.26333	18.06667	48.63333
11-01-2016 19:00	430	50	20.13333	48	19.56667	44.4	19.89	44.9	19	46.36333	17.1	55.09	6.123333	88.59	17.82333	45.49333	18.06667	48.56
11-01-2016 19:10	250	40	20.26	52.72667	19.73	45.1	19.89	45.49333	19	47.22333	17.1	55.16333	6.0675	88.215	17.96333	46.16	18.03333	48.66667
11-01-2016 19:20	100	10	20.42667	55.89333	19.85667	45.83333	20.03333	47.52667	19	48.69667	17.1	55.5	5.9	88.15667	17.96333	45.53333	18.1	49.19333
11-01-2016 19:30	100	10	20.56667	53.89333	20.03333	46.75667	20.1	48.46667	19	48.49	17.15	56.0425	5.8	88.36667	17.89	44.92667	18.15	49.2
11-01-2016 19:40	90	10	20.73	52.66	20.16667	47.22333	20.2	48.53	18.92667	48.15667	17.16667	56.49	5.726667	88.16	17.76	44.26667	18.23	49.63333
11-01-2016 19:50	70	30	20.85667	53.66	20.2	47.05667	20.2	48.4475	18.89	47.96333	17.2	56.93333	5.526667	87.3	17.7	43.72667	18.35667	50.02667
11-01-2016 20:00	80	30	20.89	51.19333	20.2	46.33	20.2	48.19333	18.96333	48.63	17.2	57.06	5.333333	86.76	17.66667	43.16	18.53333	50.2
11-01-2016 20:10	140	40	20.89	49.8	20.2	46.02667	20.16667	47.63333	19.03333	49.5	17.59333	70.72667	5.333333	87.46333	17.6	42.69333	18.66667	50.26

EXPLORATORY DATA ANALYSIS (EDA)

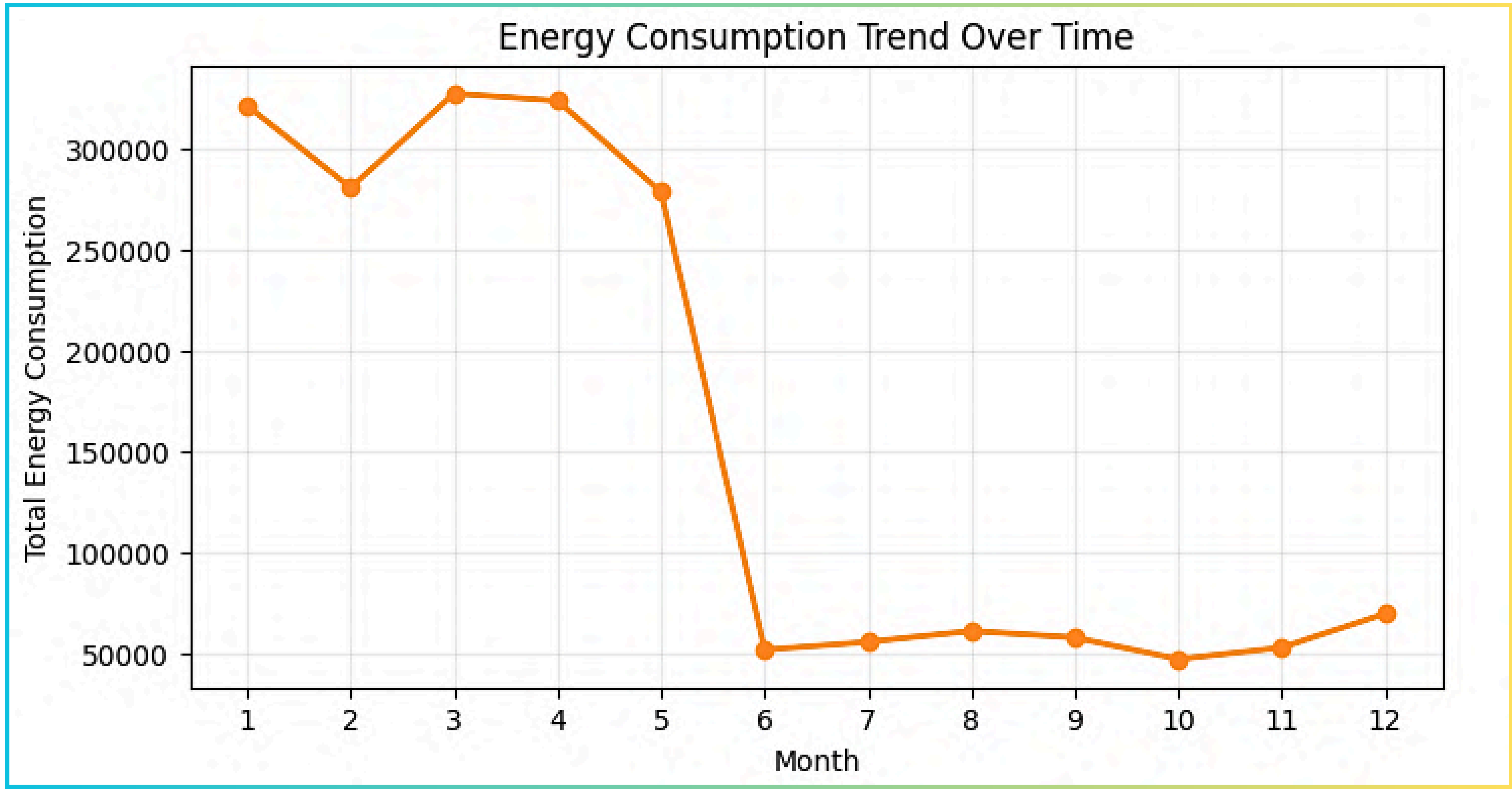
- ***Correlation analysis*** shows the relationship between features and energy consumption.
- Most features have weak correlation, indicating complex non-linear dependencies.
- “Lights” has the highest positive correlation, while “RH_out” shows a slight negative correlation.





EXPLORATORY DATA ANALYSIS (EDA)

- Energy consumption is highest from January to May, peaking around March-April.
- There is a sharp drop in consumption after May, reaching the lowest in June.
- Consumption remains relatively low in later months with a slight increase towards December.



PREPROCESSING AND FEATURE ENGINEERING

- Data preprocessing involved cleaning and preparing the dataset for modeling.
- Irrelevant or redundant features were removed, and multicollinearity was addressed.
- The dataset was then split into training and testing sets in a 80:20 ratio to evaluate model performance effectively.

Feature Engineering

- Environmental factors such as temperature and humidity trends were emphasized to improve predictive performance. e.g. : df['Temperature Inside Building']=df['T1']+df['T2']+df['T3']+df['T4']+df['T5']+df['T7']+df['T8']+df['T9'])/8
- As all the features were numeric, there was no need of encoding.

MACHINE LEARNING MODELS IMPLEMENTED

Three machine learning models were implemented in this project : Linear Regression, Random Forest, and XGBoost. These models were selected to compare performance across simple, ensemble, and boosting techniques, providing a comprehensive evaluation of prediction capabilities.

Linear Regression was used as a baseline model to establish a reference performance.

Random Forest, an ensemble method, was used to capture non-linear relationships and reduce overfitting.

XGBoost, a gradient boosting algorithm, was applied to further enhance predictive performance through sequential learning and optimization.

Hyperparameter Tuning : Utilized GridSearchCV for both Random Forest and XGBoost to optimize parameters like n_estimators, max_depth, and learning_rate.

RESULTS AND PERFORMANCE METRICS

- The performance of the models was evaluated using metrics such as R^2 score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE).
- Among all models, XGBoost performed the best, demonstrating superior ability to capture complex patterns in the data.
- Linear Regression served as a baseline, while Random Forest showed moderate performance compared to XGBoost.

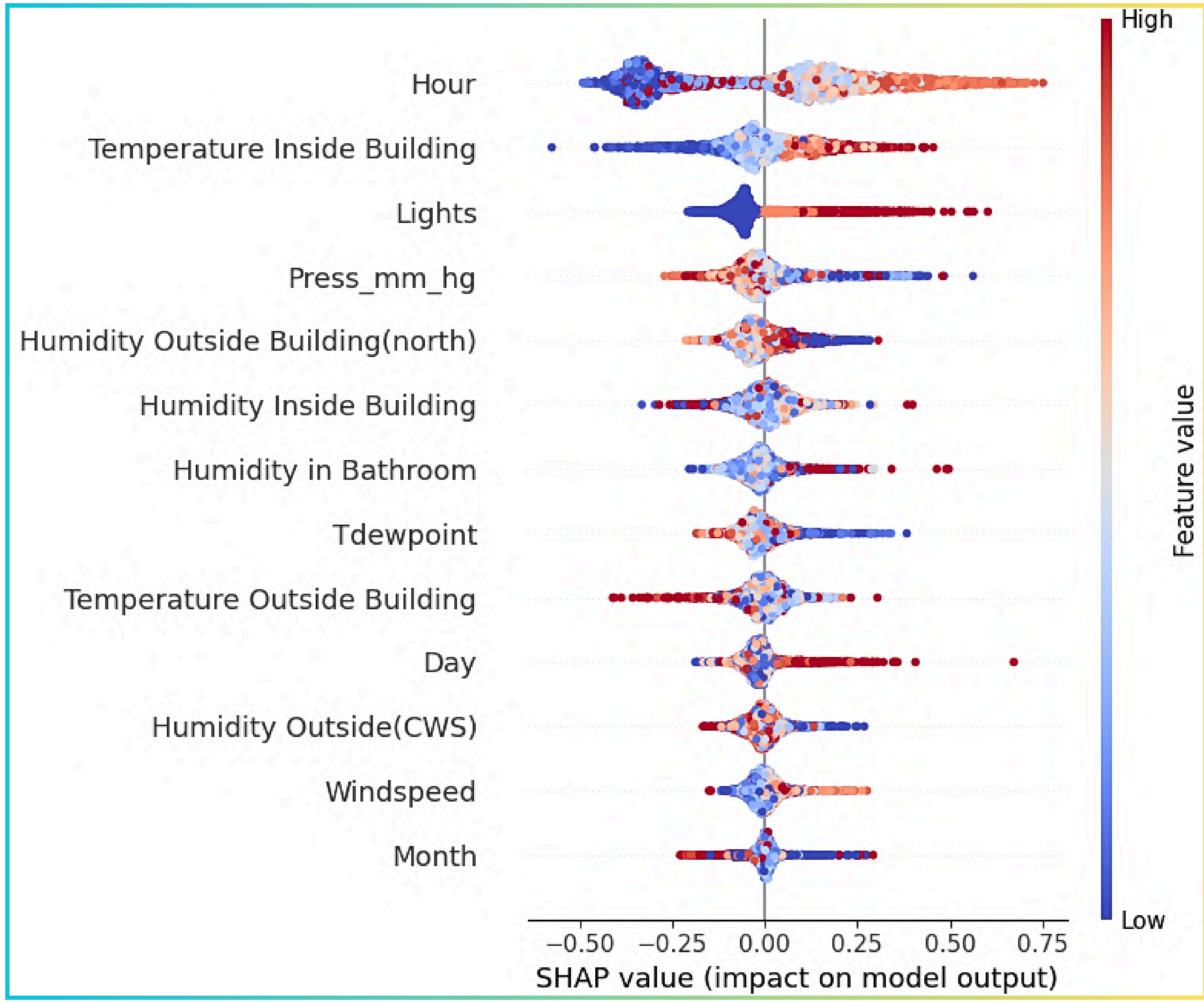
Model	Key Parameters Used	MAE	RMSE	R^2 Score
Linear Regression	Default parameters (baseline model)	0.403	0.582	18.72
Random Forest Regressor	n_estimators, bootstrap, max_depth	0.258	0.39	63.476
XGBoost Regressor	learning_rate, max_depth, min_child_weight, n_estimators	0.259	0.381	65.066

SHAP SHapley Additive exPlanations

- The SHAP plot shows that Hour, Indoor Temperature, and Lights have the strongest impact on energy consumption predictions.

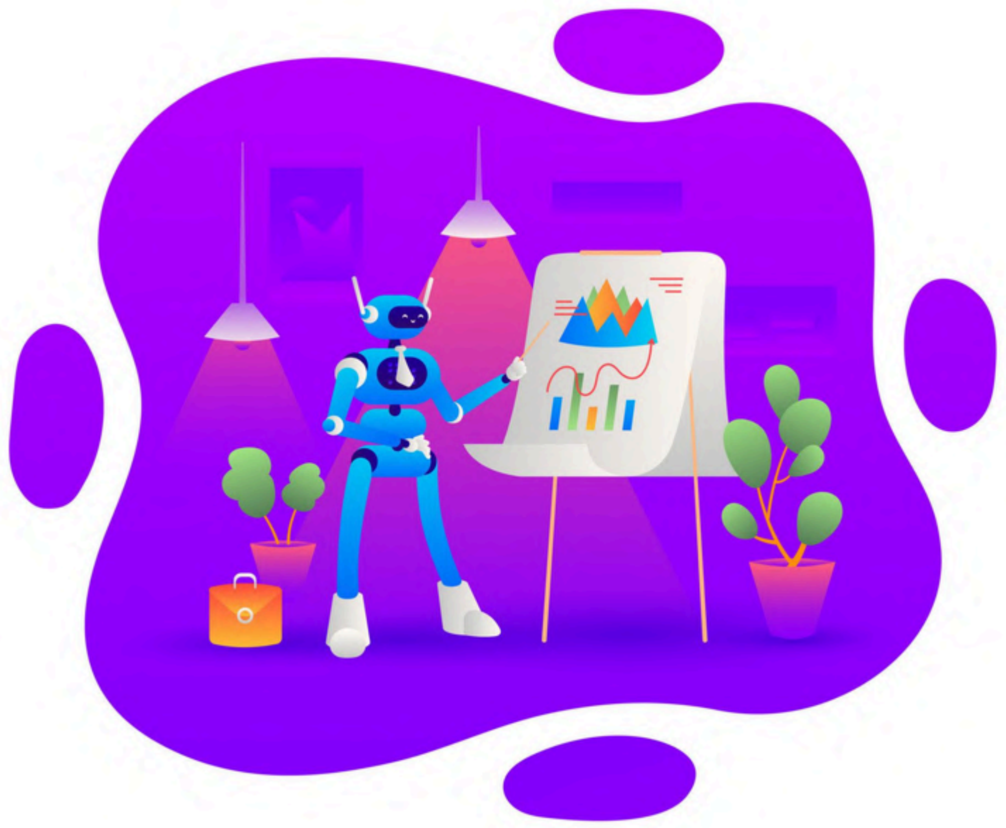
- Red (high values) generally increase energy usage, while blue (low values) decrease it.

- Most other features have smaller, mixed effects, indicating combined influence rather than a single dominant factor.



KEY INSIGHTS

- Energy depends on multiple factors.
- Individual features have weak impact.
- Combined features improve accuracy.

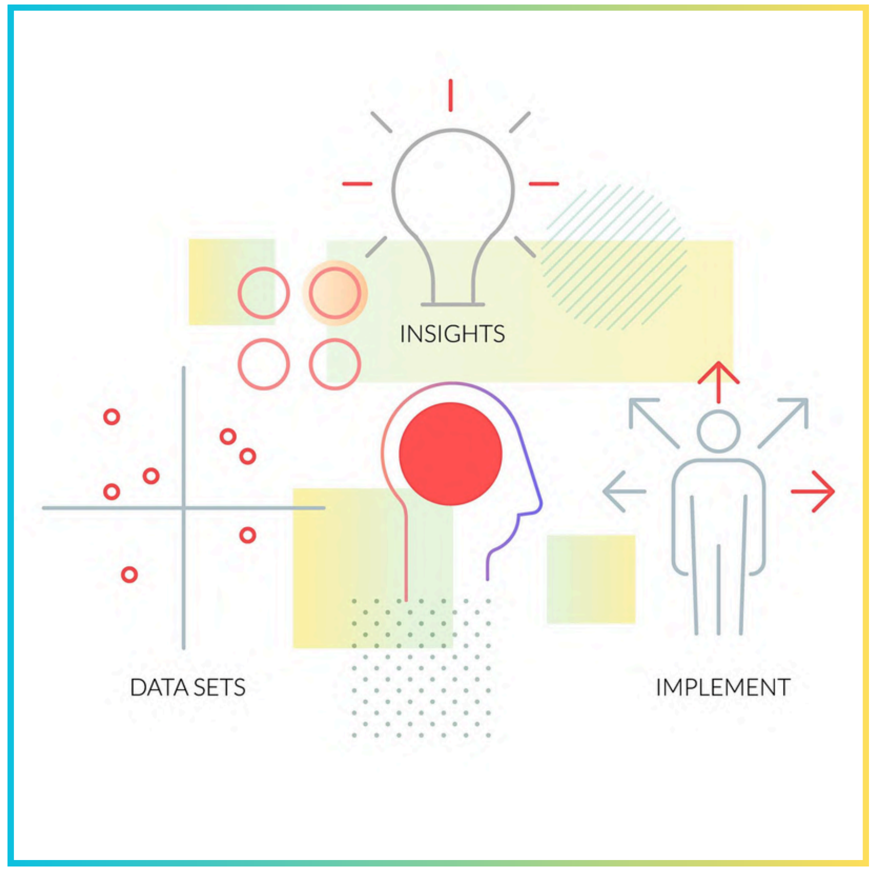


APPLICATIONS

- Smart home energy optimization.
- Cost reduction and efficiency.
- IoT-based monitoring systems.

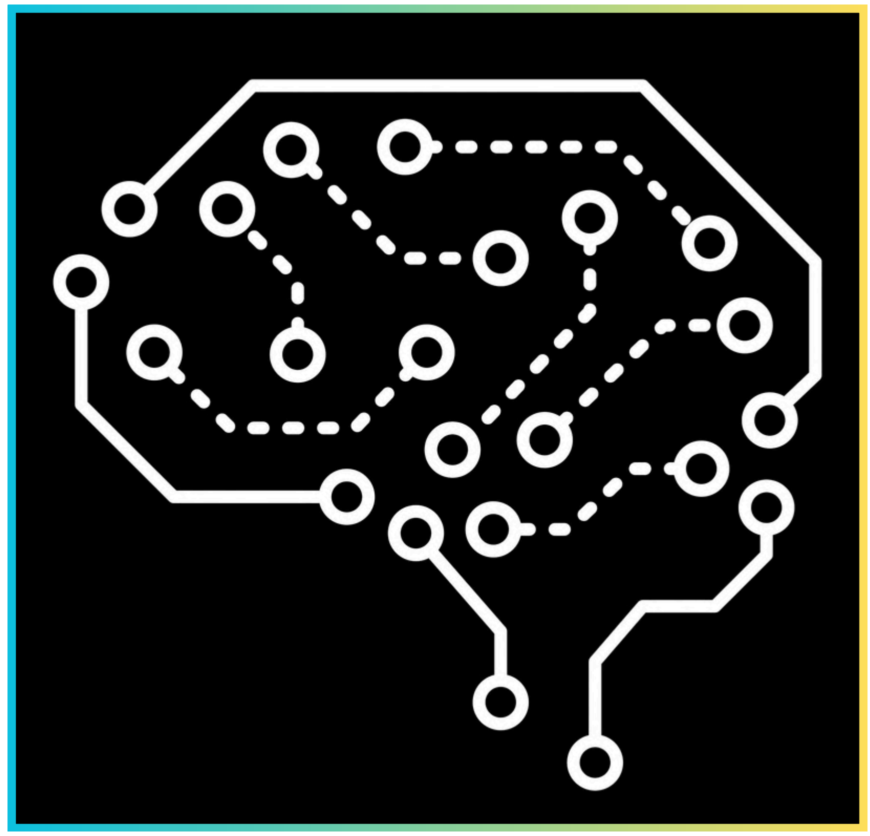
Conclusion

- ML models predict energy effectively.
- XGBoost is most accurate.
- Feature engineering improves results.



Future work

- Use deep learning models.
- Add real-time IoT data.
- Deploy as web application.



THANK YOU

Smart Homes

Energy Efficiency

Data Driven Insights

Sustainable Future